

ICG - Strategy Forge: Midterm Report Data Science

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1 Introduction to Machine Learning

Machine Learning (ML) is a subfield of Artificial Intelligence (AI) focused on building systems that learn from data and improve over time—without needing to be explicitly programmed for each specific task. Instead of following hard-coded rules, ML models identify patterns from experience and use those patterns to make predictions or decisions on new data. The real power of machine learning lies in its ability to scale: once trained, a model can process vast amounts of new data quickly and consistently.

1.1 Core Concepts

Before exploring the different learning paradigms, it helps to understand the foundational building blocks that all ML systems share:

- **Data:** The raw material. This includes input features (independent variables) and, in supervised settings, output labels (the target variable to be predicted).
- **Model:** A mathematical function that maps inputs to outputs. The model is what the algorithm “learns” during training.
- **Training:** Exposing the model to historical data so it can adjust its internal parameters to minimize error.
- **Prediction:** Applying the trained model to new, unseen data to generate outputs.
- **Evaluation:** Measuring how well the model’s predictions match the true outcomes, using metrics like accuracy or mean squared error.

1.2 Categories of Machine Learning

ML approaches are broadly divided into three paradigms based on how the model learns from data:

Type	Description	Example Use Case
Supervised Learning	Trains on labeled data where both inputs and correct outputs are provided.	Predicting house prices, spam detection
Unsupervised Learning	Finds hidden patterns or groupings in data without predefined labels.	Customer segmentation, topic modeling
Reinforcement Learning	Learns through trial and error by receiving rewards or penalties for actions.	Game-playing agents, robot navigation

2 Linear Regression

Linear Regression is one of the foundational supervised learning algorithms. It models the relationship between a dependent variable (output) and one or more independent variables (inputs) by fitting a straight line through the data. The goal is to predict a continuous numerical value—for example, estimating sales revenue based on advertising spend.

2.1 The Best Fit Line

The core idea is to find a line that best represents the trend in the data. This line minimizes the total distance between itself and all the data points—specifically, it minimizes the sum of

squared vertical distances (residuals). Once the line is established, it can be used to predict outputs for values not seen during training.

For Simple Linear Regression (one input), the line is expressed as:

$$y = mx + c \tag{1}$$

where m is the slope (rate of change) and c is the y-intercept (value of y when $x = 0$). The algorithm optimizes both values so predictions are as accurate as possible.

Interpreting the slope (m): A positive slope means the output increases as the input grows. A negative slope means they move in opposite directions. The magnitude tells you how much y changes per unit increase in x .

2.2 Variants of Linear Regression

Simple Linear Regression

Involves a single predictor variable. Best used when there is a clear, direct relationship between one input and the output. The model equation is $y = mx + c$.

Multiple Linear Regression

Extends the model to include two or more predictor variables. Each predictor has its own coefficient, and all contribute to the final prediction:

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + b \tag{2}$$

where x_1 through x_n are the input features, m_1 through m_n are their respective weights, and b is the bias term (intercept). This form is more flexible and commonly used in real-world applications where outcomes depend on multiple factors simultaneously.

3 Polynomial Regression

Not all relationships in data are linear. When data follows a curved trend—such as population growth, market saturation curves, or seasonal demand—a straight line will consistently underfit. Polynomial Regression addresses this by extending linear regression to fit curves.

It models the relationship between input x and output y as a polynomial of degree n :

$$y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n \tag{3}$$

Each coefficient b_0 through b_n is learned from the data. Higher-degree polynomials can capture more complex patterns, but also risk overfitting (memorizing noise rather than generalizing).

3.1 The Cost Function

To assess how well a regression model is performing, we need a way to quantify the error between what the model predicts and the actual observed values. This is done through the cost function.

For both linear and polynomial regression, the standard cost function is Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \tag{4}$$

where n is the total number of data points, y_i is the actual observed value, and $\hat{y}_i$ is the model's predicted value. Squaring the difference ensures all errors are positive and penalizes larger mistakes more heavily than small ones. The goal of training is to find the model parameters that minimize this value.

3.2 Slope and Intercept Calculation

For simple linear regression, the optimal slope and intercept can be derived directly from the data using closed-form expressions—no iterative optimization is required:

$$m = \frac{\sum(X - x_i)(Y - y_i)}{\sum(X - x_i)^2} \quad (5)$$

$$c = Y - mX \quad (6)$$

Here, X and Y denote the average values of the independent and dependent variables respectively. These formulas produce the exact values of m and c that minimize the MSE for a simple linear model.

4 Understanding Mean Squared Error (MSE)

Mean Squared Error (MSE) is a frequently utilized metric for assessing the precision of a regression model. It computes the average of the squared discrepancies between the predicted values and the actual observed values.

Formula:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (7)$$

- $\hat{y}_i$: Predicted value
- y_i : Actual value
- n : Number of data points

Crucial Takeaways:

- A diminished MSE (approaching 0) denotes superior predictive precision.
- Squaring the errors penalizes larger mistakes more heavily. Squaring serves two purposes: it removes negative signs (so errors don't cancel out), and it disproportionately punishes large errors—encouraging models to avoid big mistakes rather than just minimizing average deviation.
- The units of MSE are the squared units of the target variable.

5 Gradient Descent

Gradient Descent functions as an optimization technique deployed to discover the coefficient values that yield the minimum of a cost function. The fundamental concept is to begin with arbitrary values for the slope (m) and intercept (b), and subsequently iterate to adjust these values in order to systematically decrease the error. The ultimate objective is to arrive at the lowest possible point of the cost function.

5.1 The Learning Rate

The learning rate, denoted as α , acts as a tuning parameter that dictates the magnitude of the step taken toward the minimum during each iteration. Generally, α is set to a diminutive figure, such as 0.001, to guarantee that the updates remain stable and achieve proper convergence.

5.2 The Update Rule

In order to adjust the slope m , the algorithm utilizes the derivative of the cost function with respect to m . The rule for updating is defined as follows:

$$m = m - \alpha \cdot \frac{\partial}{\partial m} J(m) \quad (8)$$

Where:

- α is the learning rate
- $J(m)$ is the cost function (e.g., Mean Squared Error)
- $\frac{\partial}{\partial m} J(m)$ is the partial derivative of the cost with respect to m

This iterative procedure continues until the derivative nears zero, signaling that the minimum of the cost function has successfully been reached. The final calculated values of m (and b , if computed in a similar fashion) are subsequently applied to the best fit line equation.

6 Correlation in Machine Learning

Correlation is a statistical metric that describes the relationship between variables and is widely used in machine learning for data analysis and model improvement. The correlation coefficient ranges from -1 (strong negative relationship) to $+1$ (strong positive relationship), while a value close to 0 indicates little or no association.

Strong correlation suggests that variables tend to move together, helping uncover patterns within data.

6.1 Types of Correlation

- **Pearson Correlation Coefficient:** Measures linear relationships between continuous variables. It evaluates how closely data points follow a straight line and is independent of the measurement scale.
- **Spearman's Rank Correlation:** Spearman's coefficient (r_s) measures monotonic relationships and is useful for ordinal or non-linear data. Instead of actual values, it relies on ranked data and can capture trends where variables consistently rise or fall together.

6.2 Applications in Machine Learning

- **Feature Selection:** Correlation helps identify redundant features and reduces overfitting. Highly correlated features often provide overlapping information, so removing them can simplify models and improve generalization.
- **Bias Reduction:** Variables strongly associated with sensitive attributes such as race or gender can introduce bias. Correlation analysis helps detect and reduce such influences, promoting fairer models.
- **Multicollinearity:** When predictor variables are strongly correlated with one another, multicollinearity can increase variance and reduce interpretability. Correlation matrices and methods such as PCA can help address this issue.
- **Interpretability:** Correlation analysis highlights the variables that have the greatest impact on predictions, improving transparency and helping identify misleading relationships.

6.3 Limitations

A correlation does not establish causation. For instance, ice cream sales and shark attacks may rise together because of summer weather rather than a direct connection. Therefore, domain expertise is necessary to interpret correlations correctly.

Example: Facebook Ad Campaign

In a Facebook advertising campaign, the number of total conversions (clicks leading to the product page) showed a strong relationship with approved conversions (completed purchases), suggesting an effective conversion funnel. However, including total conversions as a feature caused overfitting. Removing this feature improved the model's ability to generalize by encouraging it to rely on other meaningful predictors.

7 Data Preprocessing

Data preprocessing is a crucial stage in the machine learning workflow. It involves cleaning, organizing, and transforming raw data into a format suitable for training models. Since real-world datasets often contain noise, inconsistencies, and missing information, preprocessing improves data quality, boosts model accuracy, and enhances the ability of models to generalize to unseen data.

7.1 Handling Missing Values

- **Removal:** Remove rows or columns containing missing values when the amount of missing data is minimal.
- **Imputation:** Fill missing entries with estimated values:
 - Mean or median for numerical variables.
 - Mode for categorical variables.

7.2 Feature Scaling

- **Standardization (Z-score Normalization):** Adjusts data so that it has a mean of zero and a standard deviation of one. Best suited for normally distributed data.

$$x' = \frac{x - \mu}{\sigma}$$

- **Min-Max Normalization:** Transforms values into a fixed range, commonly $[0, 1]$. Useful when a specific scale is required.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- **Robust Scaling:** Uses the median and interquartile range (IQR) to minimize the effect of outliers.

$$x' = \frac{x - \text{median}}{\text{IQR}}$$

7.3 Encoding Categorical Variables

- **Label Encoding:** Replaces categories with integer values (e.g., Green = 0, Yellow = 1).
- **One-Hot Encoding:** Generates separate binary columns for each category, indicating whether that category is present or absent.

8 Logistic Regression

Logistic Regression is a widely used machine learning algorithm designed for classification tasks, particularly when the target variable has two possible outcomes, such as true/false, yes/no, or 0/1. Unlike linear regression, which predicts continuous numerical values, logistic regression estimates the probability that a data point belongs to a specific class.

8.1 Mathematical Foundation

Logistic regression converts a linear combination of input features into a probability value through the sigmoid function, defined as:

$$f(t) = \frac{1}{1 + e^{-t}} \quad (9)$$

where

$$t = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n \quad (10)$$

Here:

- $x_1, x_2, \dots, x_n$ represent the input features.
- $w_1, w_2, \dots, w_n$ are the corresponding feature weights.
- w_0 denotes the bias or intercept term.

The sigmoid function compresses the output into the range $[0, 1]$, making it suitable for probability estimation.

8.2 Log Loss Cost Function

To evaluate the discrepancy between actual labels and predicted probabilities, logistic regression employs the log loss (or cross-entropy) cost function. This metric heavily penalizes predictions that are both overly confident and incorrect.

In binary classification contexts:

- y represents the true label, which is either 0 or 1.
- $p_\theta(x)$ denotes the probability predicted by the sigmoid output.

For an individual instance, the cost is calculated as $-\log(p_\theta(x))$ when $y = 1$, and $-\log(1 - p_\theta(x))$ when $y = 0$.

Across m examples in a training set, the total cost function is formulated as:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(p_\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - p_\theta(x^{(i)})) \right] \quad (11)$$

Because this function exhibits convexity, gradient descent can optimize it efficiently.

8.3 Optimizing via Gradient Descent

Gradient descent is utilized to minimize the overall cost, $J(\theta)$. The partial derivative of the cost function concerning a specific parameter θ_j is given by:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (p_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)} \quad (12)$$

Each parameter θ_j is then adjusted using this derivative:

$$\theta_j := \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j} \quad (13)$$

Regarding this update, α denotes the learning rate. This adjustment process repeats iteratively until the model reaches convergence. By continuously moving down the slope of the cost function, gradient descent shrinks the gap between actual outcomes and model predictions, ultimately finding an optimal solution that minimizes prediction errors.

9 K-Means Clustering

9.1 Overview

Clustering is a foundational unsupervised learning method designed to group data points that share similar characteristics. Thanks to its efficiency and straightforward nature, K-Means stands out as one of the most widely adopted clustering techniques.

9.2 The K-Means Algorithm

This iterative algorithm divides a dataset into K distinct groups, where each group is defined by a central point known as a centroid. The procedure operates through the following steps:

1. Begin by randomly initializing K centroids.
2. Allocate every data point to the centroid that is closest to it, measured via Euclidean distance.
3. Update the position of each centroid by calculating the mean of all data points currently assigned to that specific cluster.
4. Iterate through the assignment and recalculation steps until the centroid positions stabilize and stop changing, which indicates convergence.

9.3 Objective: Within-Cluster Sum of Squares (WCSS)

The primary goal of K-Means is to minimize the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (14)$$

In this formula:

- C_i represents the collection of data points belonging to cluster i .
- μ_i serves as the centroid for cluster i .
- $\|x - \mu_i\|^2$ is the squared Euclidean distance separating a data point x from its cluster's centroid μ_i .

9.4 Determining Optimal K Using the Elbow Method

When the actual number of natural groupings in the data is unknown, the Elbow Method acts as an essential heuristic for finding the ideal number of clusters (K) in K-Means. This technique provides both a visual and quantitative way to balance model complexity against clustering performance.

The approach gets its name from the distinctive “elbow” shape that appears when plotting the WCSS against varying K values. The elbow point signifies the optimal balance for the model. The logic behind the Elbow Method depends on WCSS, the objective function that K-Means attempts to minimize. Lower WCSS values point to cohesive, tightly packed clusters, whereas higher values indicate dispersed or poorly constructed groupings.

Implementing the Elbow Method:

1. **Data Preparation:** Select a dataset suited for clustering, typically one consisting of numerical features.
2. **Feature Selection:** Isolate the relevant features to build the feature matrix X .
3. **Initialize Variables:** Test a range of K values (for instance, 1 through 10). Train a K-Means model for each K and calculate the resulting WCSS, which is sometimes referred to as inertia.
4. **Plot the Curve:** Create a graph plotting the number of clusters (K) on the x-axis and the corresponding WCSS on the y-axis.
5. **Identify the Elbow:** Look for the K value at which the reduction in WCSS begins to level off and becomes marginal.

Prior to reaching the elbow point, adding more clusters causes a substantial drop in WCSS. Beyond the elbow point, adding further clusters yields diminishing returns.

10 Ridge and Lasso Regression

Linear regression estimates a target variable by expressing it as a combination of several predictor variables. However, when there are too many predictors or when they are highly correlated, the model may overfit the data and produce unstable coefficient estimates. To overcome these issues, Ridge and Lasso regression introduce penalty terms into the cost function, helping achieve a better balance between model accuracy and complexity.

10.1 Ridge Regression

Ridge regression applies a penalty based on the **square of the regression coefficients**. This regularization shrinks coefficient values toward zero, reducing variance and improving stability, especially when predictors are strongly correlated.

The tuning parameter λ (lambda) controls the amount of shrinkage: larger values of λ result in stronger penalties, while $\lambda = 0$ gives the ordinary linear regression model. Ridge regression has a closed-form mathematical solution, making it computationally efficient, but it does not remove variables by forcing coefficients to zero.

The Ridge Regression objective minimizes:

$$\hat{\beta}^{\text{Ridge}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (15)$$

Its closed-form solution is:

$$\hat{\beta}^{\text{Ridge}} = (X^T X + \lambda I)^{-1} X^T y \quad (16)$$

where X is the $n \times p$ design matrix, y is the response vector, and I is the identity matrix.

10.2 Lasso Regression

Lasso regression introduces a penalty based on the **absolute values of the coefficients**. This encourages sparsity by driving some coefficients exactly to zero, effectively performing feature selection. As a result, Lasso is especially useful when dealing with datasets containing many predictors.

Unlike Ridge, Lasso naturally produces simpler models with fewer selected variables. Since no direct analytical solution exists, optimization techniques such as coordinate descent are generally used.

The Lasso Regression objective minimizes:

$$\hat{\beta}^{\text{Lasso}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (17)$$

The $L1$ penalty term promotes sparse solutions by shrinking some coefficients exactly to zero.

10.3 Comparison & Applications

- **Variable Selection:** Lasso can eliminate unimportant predictors by setting their coefficients to zero, whereas Ridge only reduces their magnitude without removing them.
- **Correlated Predictors:** Ridge generally performs better when predictors are highly correlated because it distributes the coefficient values among them. Lasso may instead retain only one predictor while suppressing others.
- **Computation:** Ridge regression is easier to compute due to its closed-form solution, while Lasso relies on numerical optimization methods, making it computationally more demanding.

Ridge and Lasso regression are widely used in fields such as genomics, finance, healthcare, and text mining. Ridge is particularly effective when dealing with correlated variables, whereas Lasso is preferred when creating sparse and easily interpretable models through automatic feature selection. In situations involving numerous correlated features, **Elastic Net**, which combines Ridge and Lasso penalties, often provides improved performance.

11 Principal Component Analysis (PCA)

11.1 Important Terminologies

- **Principal Components:** These are new orthogonal directions that capture the maximum possible variance present in the data.
- **Dimensionality:** High-dimensional datasets can be difficult to analyze and visualize. PCA reduces the number of dimensions while preserving most of the useful information.
- **Correlation:** Correlation measures the relationship between variables. In PCA, the covariance matrix is examined to understand these relationships.
- **Orthogonality:** Principal components are mutually orthogonal (uncorrelated), ensuring that no redundant information is shared between them. Mathematically, $\sum_{k=1}^p a_{ik} a_{jk} = 0$.
- **Eigenvectors:** These define the directions along which the data shows maximum variation and are obtained from the covariance matrix.
- **Eigenvalues:** They indicate the amount of variance explained by their corresponding eigenvectors.

- **Covariance Matrix:** For a dataset represented by matrix X with n observations and p variables, the covariance matrix is $C = \frac{1}{n}X^T X$, where X^T denotes the transpose of X .

11.2 Steps of the PCA Algorithm

1. **Standardize the data** so that each feature has a mean of zero and a standard deviation of one.
2. **Compute the covariance matrix** to measure relationships among variables.
3. **Determine eigenvectors and eigenvalues**, which identify the principal directions and quantify their importance.
4. **Select the principal components** by choosing those associated with the largest eigenvalues, preserving most of the dataset's variance.
5. **Transform the original data** by projecting it onto the selected lower-dimensional feature space.

11.3 Applications of PCA in Machine Learning

- Visualizing datasets with many dimensions.
- Reducing the number of features in healthcare and biomedical data.
- Performing image compression and resizing.
- Studying financial datasets and identifying market trends.
- Detecting hidden patterns and structures in large-scale datasets.